

J. Alex Baecher

Curriculum Vitae

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EDUCATION

Doctor of Philosophy, Interdisciplinary Ecology <i>University of Florida</i>	Aug 2024
Master of Science, Biology <i>Eastern Kentucky University</i>	Dec 2017
Bachelor of Science, Biology <i>University of Arkansas</i>	May 2014

APPOINTMENTS

Assistant Professor of Applied Ecology <i>Emory University</i> Department of Environmental Science	Aug 2026 – Present
Postdoctoral Fellow in Quantitative Ecology <i>Utah State University</i> Department of Rangeland Resources	July 2025 – July 2026
Postdoctoral Research Associate <i>University of Florida</i> Institute for Food and Agricultural Science	Aug 2024 – July 2025
Doctoral Preeminence Fellow <i>University of Florida</i> Department of Wildlife Ecology	Aug 2019 – Aug 2024
Conservation and Research Technician <i>Memphis Zoo</i> Department of Conservation and Research	Jan 2018 – July 2019
NSF REU Research Coordinator <i>Eastern Kentucky University</i> Department of Biology	May 2017 – Aug 2017
Graduate Teaching Assistant <i>Eastern Kentucky University</i> Department of Biology	May 2015 – Aug 2017
Research Assistant <i>University of Arkansas</i> Department of Biology	Mar 2012 – Mar 2015

PUBLICATIONS

Reverse chronological • §Mentee co-author • Citations: 438 • h-index: 8

Published

2026

17. §Akshay, VA, **JA Baecher**, N Burkett-Cadena, J Thorson, §Y Sánchez, §Y Tavares, §A Bauer, RP Guralnick, LP Campbell. **2026**. *Linking hosts, landscapes, and climate to advance zoonotic arbovirus forecasting*. *Scientific Reports* [[html](#)] [[code](#)] [[preprint](#)]
16. Oliveira, BF, et al. [+21 co-authors including **JA Baecher**]. **2026**. *Species range shifts often speed ahead of their climatic niches*. *Proceedings of the National Academy of Sciences* [[html](#)] [[preprint](#)] [[code](#)]
Press: [[Nature EcoEvo](#)] [[Conservation Science Partners](#)]

2025

15. **Baecher, JA**, §VA Akshay, RP Guralnick, §AM Bauer, §Y Tavares, §Y Sánchez, LP Campbell. **2025**. *Toward ecological forecasting of West Nile virus in Florida: Insights from two decades of sentinel chicken monitoring*. *Science of the Total Environment* 1000:180308 [[html](#)] [[pdf](#)] [[code](#)]
Press: [[Florida Museum](#)] [[The Microbiologist](#)]

2024

14. Evans, LJ, **JA Baecher**, BR Scheffers. **2024**. *Invasion risk posed by the pet trade*. *Frontiers in Ecology and the Environment* [[html](#)] [[pdf](#)] [[data](#)]
13. **Baecher, JA**. **2024**. *The environmental filtering paradigm and non-filtering community assembly processes*. *Journal of Biogeography* 51(11):2109–2116 [[html](#)] [[pdf](#)]
12. Klinges, DH, **JA Baecher**, JJ Lembrechts, IMD Maclean, J Lenoir, C Greiser, M Ashcroft, LJ Evans, MR Kearney, J Aalto, LC Barrio, P De Frenne, J Guillemot, K Hylander, T Jucker, M Kopecky, M Luoto, M Macek, I Nijs, J Urban, L van den Brink, P Vangansbeke, J Von Oppen, J Wild, J Boike, R Canessa, M Noretto, A Rubtsov, J Sallo-Bravo, BR Scheffers. **2024**. *Proximal microclimate: moving beyond spatiotemporal scale to improve predicted biological responses*. *Global Ecology and Biogeography* 33(7):e1–16 [[html](#)] [[pdf](#)] [[code](#)]
Award: [[Wiley's Top 10 Most Cited](#)]
11. Comte, L, R Bertrand, SE Diamond, LT Lancaster, ML Pinsky, BR Scheffers, **JA Baecher**, RMWJ Bandara, I-C Chen, JA Lawlor, NA Moore, BR Oliveira, J Murienne, J Rolland, MA Rubenstein, J Sunday, LM Thompson, F Villalobos, SR Weiskopf, J Lenoir. **2024**. *Bringing traits back in the equation: A roadmap to understand species redistribution*. *Global Change Biology* 30:e17271 [[html](#)] [[pdf](#)]
10. Lawlor, JA, L Comte, G Grenouillet, J Lenoir, **JA Baecher**, RMWJ Bandara, R Bertrand, I-C Chen, SE Diamond, LT Lancaster, NA Moore, J Murienne, BF Oliveria, GT Pecl, ML Pinsky, J Rolland, MA Rubenstein, BR Scheffers, LM Thompson, B can Amerom, F Villalobos, SR Weiskopf, J Sunday. **2024**. *Mechanisms, detections, and impacts of species redistributions under climate change*. *Nature Reviews Earth & Environment* 5:351–368 [[html](#)] [[pdf](#)]

Impact Factor: 71.5 Altmetric: 134 (top 5%)

Press: [[list of articles](#)]

2023

9. **Baecher, JA**, SA Johnson, EA Roznik, BR Scheffers. **2023**. *Experimental evaluation of how invasions and climate change interact to alter the vertical assembly of an amphibian community. **Journal of Animal Ecology** 92(4):875–888* [[html](#)] [[pdf](#)] [[code](#)]
Press: [[Nature Climate Change](#)] [[Florida Today](#)] [[Nature Climate Change](#)]

2022

8. Basham, EW, **JA Baecher**, DH Klings, BR Scheffers. **2022**. *Vertical stratification patterns of tropical forest vertebrates: A meta-analysis. **Biological Reviews** 98(1):99–114* [[html](#)] [[pdf](#)] [[code](#)]
7. Basham, EW, M González-Pinzón, A Romero-Marcucci, N Carl, **JA Baecher**, BR Scheffers. **2022**. *Large, old trees define the vertical, horizontal, and seasonal distributions of a poison frog. **Oecologia** 199:257–269* [[html](#)] [[pdf](#)] [[code](#)]

2020

6. Swartwout, M, PN Vogrinc, **JA Baecher**, CS Kross, JD Willson. **2020**. *Prey Size and Growth Rate do not Influence Trophic Morphology of Juvenile Water Snakes Nerodia sipedon. **Herpetologica** 76(1):53–60* [[html](#)] [[pdf](#)]
5. **Baecher, JA**, SE Jones, SC Richter. **2020**. *Tail bifurcation in two species of Desmognathus salamander (Caudata: Plethodontidae) in southeastern Kentucky, USA. **Herpetological Bulletin** 150:29–30* [[html](#)] [[pdf](#)]

2019

4. Hinkson, KM, **JA Baecher**, S Poo. **2019**. *Cryopreservation and hormonal induction of spermiic urine in the smooth-sided toad Rhaebo guttatus. **Cryobiology** 89:109–111* [[html](#)] [[pdf](#)]
Press: [[The Commercial Appeal](#)] [[USA Today](#)]

2018

3. **Baecher, JA**, SC Richter. **2018**. *Environmental gradients in an old-growth Appalachian Forest predict fine-scale distribution, co-occurrence, and abundance of woodland salamanders. **Ecology and Evolution** 8(24):12940–12952* [[html](#)] [[pdf](#)] [[code](#)]

2017

2. **Baecher, JA**, PH Vogrinc, JC Guzy, CS Kross, JD Willson. **2017**. *Herpetofaunal Communities in Restored and Unrestored Remnant Tallgrass Prairie and Associated Wetlands in Northwest Arkansas. **Wetlands** 38:157–168* [[html](#)] [[pdf](#)]

2014

1. **Baecher, JA**, PH Vogrinc, JC Guzy, JC Neal, JD Willson. **2014**. *Lithobates areolatus, Predation. **Herpetological Review** 45(4):681–682* [[pdf](#)]

Forthcoming

6. **Baecher, JA**, D Klinges, LJ Evans, CA Romagosa, R Fletcher, BR Scheffers. *In revision*. *Isolating what limits the spread of an invasive predator*. **Journal of Applied Ecology** [code] [preprint]
5. Thompson, LM, et al. [+21 co-authors including **JA Baecher**]. *In review*. *The RAD (Resist-Accept-Direct) management of species redistribution under climate change*. **Frontiers in Ecology and the Environment**
4. Klinges, DH, **JA Baecher**, L Soifer, BR Scheffers. *In review*. *Land modification defines climate change as experienced by life on earth*. **Science**
3. Pittman, C, et al., [+12 co-authors including **JA Baecher**]. *Submitted*. *Connecting living and preserved collections: Standardizing and integrating biological data to cultivate information sharing*. **Biodiversity Informatics**
2. **Baecher, JA**, D Klinges, LJ Evans, BR Scheffers. *In prep*. *Informing species range-shifts with global climate and habitat connectivity*. Prepared for **One Earth**
1. **Baecher, JA**, E Stuber, TJ Clark-Wolf. *In prep*. *Mapping range shifts with a novel integrated spatiotemporal N-mixture Bayesian model*. **Methods in Ecology and Evolution**

FUNDING

Reverse chronological

Amount	Organization	Fund Name	Year
\$680,000	Emory University	Lab Start-up Fund	2026
\$130,000	US Geological Survey	Climate Adaptation Postdoctoral Fellows	2025
\$195,000	University of Florida	Doctoral Preeminence Fellowship Award	2019
\$4,000	University of Florida	Robin Nadeau Graduate Research Award	2019
\$5,000	National Science Foundation	Research Experience for Undergraduates	2013
\$5,000	National Science Foundation	Research Experience for Undergraduates	2012
\$1,100	Eastern Kentucky University	Marcia Athey Research Grant	2016
\$800	Eastern Kentucky University	Dept. Biology Travel Grant	2017
\$700	Society Freshwater Science	Instars Fellowship	2014
\$700	Society Freshwater Science	Instars Fellowship	2013
\$500	Soc. Study Amphibians & Reptiles	Grant-in-Aid of Herpetological Research	2016
\$500	Eastern Kentucky University	Dep. Biological Sciences Travel Stipend	2016
\$500	Eastern Kentucky University	Div. Natural Areas Research Grant	2016
\$500	University of Arkansas	Dept. Biology Travel Grant	2013

Total Funding: \$1,024,000

PROFESSIONAL ACTIVITIES

Peer Review

Total Requested: 38 | **Total Reviewed:** 25

Biological Conservation, Conservation Science and Practice, Ecography, Ecology, Ecology & Evolution, Frontiers in Ecology and Environment, Herpetological Bulletin, Herpetological Conservation Biology, Herpetological Review, Hydrobiologia, Natural Areas Journal, Nature Ecology & Evolution, Oecologia, Scientific Reports, Wildlife Society Bulletin

Professional Membership

<i>Ecological Society of America</i>	2019 – Present
<i>International Association of Landscape Ecologists – North America</i>	2020 – Present
<i>Climate Adaptation Science Center (SE & SW)</i>	2019 – Present
<i>Southeast RISCC</i>	2022 – Present
<i>American Society of Ichthyologists and Herpetologists</i>	2014 – 2021
<i>Southeastern Partners in Amphibian and Reptile Conservation</i>	2014 – 2021
<i>Society of Freshwater Ecology</i>	2012 – 2014

Working Groups

<i>BioShifts</i> (Member, Group Leader)	2022 – Present
<i>DirtClim</i> (Core team)	2019 – Present
<i>National Synthesis for Ecological Analysis and Synthesis</i> (Member)	2025 – Present
<i>ResearchR</i> (President of Scientific Integrity)	2025 – Present
<i>ZooMu Network</i> (Member, Group Leader)	2024 – Present
<i>USGS Climate Adaptation Science Center</i> (Postdoctoral Fellow)	2025 – Present

SELECT PRESENTATIONS

Reverse chronological • †Student mentee presenter • Total presentations: 41

12. **Baecher, JA. 2026.** *Ecology of rapid change: From behavior to movements to range dynamics.* **Emory University**, Atlanta, GA [[slides](#)] [[video](#)]
11. **Baecher, JA. 2026.** *Spatial structures for ecological models: A tutorial for point-referenced data.* **Utah State University** (hybrid) [[slides](#)] [[online resource](#)]
10. **Baecher, JA. 2025.** *Shifting Frontiers: Understanding species abundance range-shifts for informed conservation strategies.* **Pacific Flyway Partners** (hybrid) [[slides](#)]
9. **Baecher, JA. 2024.** *Influence of Global Change on Species' Behavior, Movement, and Ranges.* **SNRE Seminar Series**, Gainesville, FL [[flyer](#)]
8. **Baecher, JA, L Evans, BR Scheffers. 2024.** *Toward knowledge co-production in invasion modeling: Developing novel interactive expert-elicitation software.* **Invasive Species Research Institute Conference**, Gainesville, FL [[slides](#)]
7. **Baecher, JA, D Klings, L Evans, BR Scheffers. 2023.** *Informing Species Range-shifts with Global Connectivity.* **Species On The Move**, Ft Myers, FL [[slides](#)]

6. **Baecher, JA**, L Evans, BR Scheffers. **2022**. *Predicting the spread of invasive Burmese pythons in Florida*. **SERISCC Network Webinar** (virtual) [[slides](#)] [[video](#)]
5. **Baecher, JA**, SB Reichling. **2018**. *Current status of the Louisiana Pinesnake captive breeding program at the Memphis Zoo*. **USFWS Working Group Meeting**, Shreveport, LA
4. **Baecher, JA**, SC Richter. **2018**. *Natural gradients influence the fine-scale distribution and abundance of salamanders in an old growth Appalachian forest*. **Joint Meeting of Ichthyology and Herpetology**, Rochester, NY
3. **Baecher, JA**, P Vongrinc, JD Willson. **2016**. *Herpetofauna of degraded and restored remnant Tallgrass Prairie*. **Joint Meeting of Ichthyology and Herpetology**, New Orleans, LA
2. **Baecher, JA**, DR Leasure, D Lynch, DD Magoulick. **2013**. *Impact of anthropogenic land use and hydrologic disturbance on crayfish assemblages*. **Joint Meeting of Aquatic Sciences**. Portland, OR
1. **Baecher, JA**, J Ortega, SJ Beaupré. **2013**. *Use of stable isotope analysis for determining the diet breadth of the Western Cottonmouth (Agkistrodon Piscivorous)*. **Society for Freshwater Science Annual Meeting**, Jacksonville, FL

[[see all presentations](#)]

WORKSHOPS

Reverse chronological

1. Fletcher, RJ, **JA Baecher**, M Lezzi, A Marx. **2023**. *Application of connectivity modeling to identify and predict movement and range redistribution*. **Species on the Move**, Bonita Springs, FL [[slides](#)]
2. **Baecher, JA**, L Evans, BR Scheffers. **2022**. *Expert elicitation of key invasive species habitat requirements, prevalence, and life history attributes*. **Austin Cary Forest**, Gainesville, FL [[slides](#)]
3. Fletcher, RJ, **JA Baecher**, M Lezzi, A Marx. **2022**. *Predicting the spread of invasive Burmese pythons in Florida*. **Annual Meeting of the International Association of Landscape Ecologists - North America** (virtual) [[video](#)]

TEACHING

Instructor of Record

University of Florida | School of Natural Resources and Environment
Environmental Science Lab

Fall 2022 & 2023

Teaching Assistant

Eastern Kentucky University | Department of Biology
Human Anatomy Lab

Spring & Fall 2015 – 2017

Guest Lectures

Utah State University Department of Wildland Resources Landscape Ecology	Fall 2025
University of Florida School of Natural Resources and Environment Community Science for Ecologists	Spring 2025
University of Florida School of Natural Resources and Environment Ecology of Climate Change	Spring 2023
University of Florida School of Natural Resources and Environment Wildlife of Florida	Spring 2022
Rhodes College Department of Biology Vertebrate Biology	Spring 2023
Eastern Kentucky University Department of Biology Applied Learning in Biology	Spring 2016
Radford University Department of Biology Appalachian Herpetology	Spring 2016
University of Arkansas Department of Biology Herpetology Herpetology	Spring 2014

STUDENT MENTORSHIP

Graduate student (MS, PhD) • Φ **published** • Ψ **conference** • Ω **committee member** • $\rightarrow$ **career prep**

Eastern Kentucky University** 2015 – 2017
Alex Gibson, Emily Jones (Φ , Ψ , $\rightarrow$ MS), Sarah Ashbrook (Ψ), Peyton Blair (Ψ), [[Peter Grap](#)] (Ψ , $\rightarrow$ MS, $\rightarrow$ PhD), Kelley Hoefer (Ψ), Myesha James (Ψ), [[Jacob Peters](#)] (Ψ , $\rightarrow$ MS, $\rightarrow$ PhD), [[Julia Portmann](#)] (Ψ , $\rightarrow$ MS, $\rightarrow$ PhD), [[Jocelyn Stalker](#)] (Φ , Ψ , $\rightarrow$ MS, $\rightarrow$ PhD)

Rhodes College 2018 - 2019
[[Madeline Estes](#)] (Ψ , Ω , $\rightarrow$ MS)

University of Florida 2019 – 2025
Brittany Arthur, Noah Bowen, Sierra Scauzillo (Ψ , Ω), [[V Anand Akshay](#)] (Φ , Ψ , PhD), [[Amy Bauer](#)] (Φ , Ψ , PhD), [[Yasmin Tavares](#)] (MS, Φ , Ψ , $\rightarrow$ PhD), [[Yesenia Sanchez](#)] (Φ , Ψ)

Utah State University 2025 – Present
Jack Rasmussen (PhD), [[Mallory Sandoval-Lambert](#)] (MS), [[Adrien Bastidas](#)] (PhD)

PUBLIC RESOURCES

Available at [questlab.dev]

<i>Mixed Effects Models in R</i> 	2025
<i>Spatial Structures in Ecological Models</i> Tutorial 	2025
<i>Intro to R Coding:</i>	
Part 1: the basics 	2022
Part 2: advanced techniques 	2022
<i>Machine learning the “Tidy” way</i> 	2021
<i>Linear regression with gradient decent</i> 	2021
<i>Mapping ecological flow in R</i> 	2020
<i>Empirical Dynamic Models for forecasting</i> 	2016

NON-FUNDED PROPOSALS

Proposals available upon request

Amount	Organization	Decision	Year
\$235,000	NSF: Postdoctoral Research Fellowship – Biology	Meritorious	2024
\$235,000	NSF: Postdoctoral Research Fellowship – Biology	Highly Meritorious	2023
\$25,000	The Wilderness Society: Gloria Barron Fellowship	Declined	2021
\$3,500	American Museum Natural History: Roosevelt Memorial Grant	Declined	2020
\$4,000	Foundation for the Conservation of Salamanders	Declined	2020
\$60,000	Disney Conservation Grant	Inquiry declined	2019
\$10,000	National Park Service: Tanner Memorial Fellowship	Inquiry declined	2019

Total Non-funded: \$572,500

REFERENCES

Since PhD (2019 - Present)

Robert Guralnick, PhD	2019 – Present
Curator of Biodiversity Informatics Florida Natural History Museum 352-273-1980 • [email] • [webpage]	
Brett Scheffers, PhD	2019 – Present
Associate Professor Dept of Wildlife Ecology, University of Florida Director Florida Climate Institute, University of Florida 352-846-0570 • [email] • [webpage]	
Lindsay Campbell, PhD	2024 – Present
Assistant Professor Florida Medical Entomology Lab, University of Florida 772-226-6666 • [email] • [webpage]	
TJ Clark-Wolf, PhD	2025 – Present
Assistant Professor Dept of Wildland Resources & Ecology Center, Utah State University email preferred • [email] • [webpage]	
Erica Stuber, PhD	2025 – Present
Assistant Unit Leader - USGS UT Cooperative Fish and Wildlife Research Unit Assistant Professor - Dept. of Wildland Resources & Ecology Center email preferred • [email] • [webpage]	